PROJECT spacetimes — README

A singlet superfluid of a 16-component Dirac field on 4+4 and on 7+1 Minkowski spacetime: where everything is, and how to check it

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November 12, 2026

1. What this project is

The project asks one question for two flat eight-dimensional spacetimes, one with four space-like and four time-like dimensions (4+4) and one with seven space-like dimensions and one time-like dimension (7+1). Can a 16-component Dirac field form a spin-singlet superfluid whose pairing is mediated by a new scalar boson, the *Vorton-Scalar*? And can that superfluid be the Higgs field, the dark matter, or the remedy of the Hubble tension?

The coordinates are $x0, \ldots, x7$ in both spacetimes: $x0$ is a hidden space-like coordinate, $x1, x2, x3$ are ordinary space, $x4$ is ordinary time, and $x5, x6, x7$ are additional coordinates (time-like in 4+4, space-like in 7+1).

2. The answers in brief

The complete statements, with their proofs, their numbers and their limits, are in the two manuscripts; each section of the manuscripts ends with a subsection "What has been established, and what has not".

- **The matrices.** Each spacetime has eight *real* 16×16 Dirac matrices. Their anticommutation relations, and the fact that their scaled commutators generate the spin groups (with the matrices themselves: $\mathrm{Pin}(4,4)$ and $\mathrm{Pin}(7,1)$), are proved and verified by 106 machine checks and by an independent second program.
- **The superfluid.** Established within stated approximations: the field has a pair of spin 0 that is a scalar of ordinary spacetime; a complex scalar field, the Vorton-Scalar, binds this pair, and its neutral current selects it; at any finite density the paired state lies below the unpaired one. The singlet along the hidden coordinate is a ground state, but one of a degenerate family.
- **The Higgs hypothesis.** Not established. On 4+4 it has no geometric basis. On 7+1 one structural condition is met (the order parameter is a multiplet of the rotations of the four extra axes, and its vacuum value breaks $SO(4)$ to $SO(3)$, the pattern of the Higgs vacuum), but the dynamical conditions are not met where the model can be solved. In both spacetimes a condensate in empty space was found only at couplings at which the methods used are a guide and not a proof.
- **The Hubble tension.** Not explained: simulations with a purpose-built Einstein-Boltzmann solver show that the dark radiation of the model raises the Hubble constant only at a price that the data do not allow.
- **Dark matter.** A superfluid that could act on galactic scales erases the observed small-scale structure. Heavy bound pairs as cold dark matter are neither supported nor excluded, and make no use of the superfluid.

3. Where to start reading

If you want	Read
the whole argument for one spacetime	`manuscript/16 Component Dirac Fermion Condensate Singlet Superfluid on 4+4 Spacetime.pdf`, or `... on 7+1 Spacetime.pdf`
the proofs about the matrices	`dirac matrices.pdf`
the Vorton-Scalar and the superfluid in detail	`supplement/S1_vorton_scalar_and_superfluid.pdf`
the Higgs hypothesis	`supplement/S2_higgs_hypothesis.pdf`
the cosmological simulations	`supplement/S3_cosmology.pdf`
the effective action of Cornwall, Jackiw and Tomboulis	`supplement/S4_cjt_effective_action.pdf`
what every symbol means, and how it differs from the symbols of the original files	`provenance/P1_master_symbols.pdf`
how to compile and run the code	`provenance/P8_code.pdf`

Every document exists in three versions with the same name: Markdown (`.md`), LaTeX (`.tex`) and PDF (`.pdf`).

4. What is in the repository

Path	Content
`manuscript/`	the two manuscripts (REVTeX, PDF, Markdown) and their bibliography
`dirac matrices.*`	provenance “dirac matrices”: definitions, theorems, proofs and worked examples for both sets of matrices
`supplement/`	Supplements S1 to S4
`provenance/`	P0 (how commits reach GitHub), P1 (MASTER provenance of the symbols), P2 (the PDF files used), P3 (the notebooks used), P8 (the code), P9 (the error log); `symbols.csv`; `inputs_sha256.txt`; `generated/` (fragments made by the programs)
`code/spacetimes-rs/`	the Rust program that computes everything (no external libraries; 63 unit tests)
`code/scripts/`	Python scripts: independent verification, tables, figures, building of the documents, checks of the symbols
`tools/`	reading the Mathematica notebooks without Mathematica
`data/`	every data file that a program wrote, each with a header that states the program, the command, the content and the units
`figures/`	every figure, as PDF and PNG
`docs_src/`	the sources of the documents, with placeholders for the generated numbers

Not in the repository: the author’s Mathematica notebooks, the nine published papers of the project folder, and the task specification. The provenances identify these files by their checksums (`provenance/inputs_sha256.txt`).

5. How to check the results

The short form (details and explanations: `provenance/P8_code`):

```
cd "/Users/nsh/Developer/github/spacetimes/code/spacetimes-rs"
cargo test --release                          # 63 unit tests
cd "/Users/nsh/Developer/github/spacetimes"
code/spacetimes-rs/target/release/spacetimes matrices          # the 106 machine checks
python3 code/scripts/verify_matrices_independent.py            # the independent program
python3 code/scripts/check_symbols.py                          # the checks of the symbols
```

The first command needs the Rust compiler; `provenance/P8_code` says how to install it and how to recompute every data file, table and figure.

6. Rules that the project follows

- Every object has one symbol; symbols of the 4+4 spacetime carry the tag 1644, those of the 7+1 spacetime the tag 1671, and those of the Vorton-Scalar the suffix VS (`provenance/P1_master_symbols`).
- No number in a document is typed by hand: programs write data files, and scripts insert the numbers into the documents.
- Every error that was found is recorded, with the way it was found and its correction (`provenance/P9_error_log`).
- What is not established is said, in every section.

7. Licence

The source code is distributed under the GNU General Public License, version 3 or later (file `LICENSE`).